Cambridge Assessment International Education

# Cambridge O Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

**CHEMISTRY** **5070/22**

Paper 2 Theory **May/June 2023**

**1 hour 45 minutes**

You must answer on the question paper.

No additional materials are needed.

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

**INFORMATION**

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].
- The Periodic Table is printed in the question paper.

This document has **20** pages. Any blank pages are indicated.


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**1** Choose from the following oxides to answer the questions.

**CO**
**$CO_2$**
**CaO**
**CuO**
**$Fe_2O_3$**
**$H_2O$**
**$SO_2$**
**$SiO_2$**
**ZnO**

Each oxide may be used once, more than once or not at all.

State which oxide:

**(a)** is neutral

..... [1]

**(b)** reacts with calcium oxide to form slag in the blast furnace

..... [1]

**(c)** reacts with warm dilute hydrochloric acid to give a blue coloured solution

..... [1]

**(d)** is amphoteric

..... [1]

**(e)** contains an ion with an oxidation number of +3

..... [1]

**(f)** decolourises acidified aqueous potassium manganate(VII).

..... [1]

[Total: 6]

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**2** Group VII includes the elements fluorine, chlorine, bromine and iodine.

**(a)** Chlorine is a green gas at room temperature and pressure.

State the appearance of iodine at room temperature and pressure.

..... [2]

**(b)** Chlorine reacts with aqueous potassium iodide in a displacement reaction.

$$Cl_2(aq) + 2I^-(aq) \longrightarrow 2Cl^-(aq) + I_2(aq)$$

**(i)** Explain, in terms of electrons, why chlorine is an oxidising agent in this reaction.

.....

..... [1]

**(ii)** State the oxidation number of iodine in $I_2$.

..... [1]

**(iii)** Describe what is observed during the displacement reaction.

..... [1]

**(c)** The rate of diffusion of fluorine gas is greater than that of chlorine gas under the same conditions of temperature and pressure.

**(i)** State what is meant by the term diffusion.

.....

.....

..... [1]

**(ii)** Explain why the rate of diffusion of fluorine is greater than that of chlorine under the same conditions.

.....

.....

..... [1]

**(iii)** The rate of diffusion of fluorine increases as the temperature increases.

Suggest why using ideas about kinetic particle theory.

.....

..... [1]

[Total: 8]

**3** This question is about the preparation of salts.

**(a)** Zinc nitrate is a soluble salt.

It is prepared by the reaction of an insoluble carbonate with a dilute acid.

The equation for this reaction is shown.

$$ZnCO_3(s) + 2HNO_3(aq) \rightarrow Zn(NO_3)_2(aq) + H_2O(l) + CO_2(g)$$

A sample of 4.50 g of zinc carbonate is added to 50.0 $cm^3$ of 1.30 mol / $dm^3$ nitric acid.

**(i)** Show by calculation that the zinc carbonate is in excess.

[3]

**(ii)** Once the reaction has finished the mixture is filtered.

State why the mixture is filtered.

.....

..... [1]

**(iii)** Describe how to make pure, dry zinc nitrate crystals from an aqueous solution of zinc nitrate.

.....

.....

.....

.....

..... [3]

**(b)** Lead chloride is an insoluble salt.

It is prepared using a precipitation reaction.

Name **two** aqueous solutions that react together to give a precipitate of lead chloride.

..... and ..... [1]

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**(c)** Ammonium sulfate is a soluble salt.

It is prepared by the reaction of an alkali and an acid.

Name the alkali and the acid used.

alkali .....

acid .....

[1]

[Total: 9]

**4** This question is about compounds that contain magnesium and nitrogen.

**(a)** The formula for a nitride ion can be written as $^{15}_{7}N^{3-}$.

Complete Table 4.1 to show the number of particles in this nitride ion.

**Table 4.1**

| particle | number of particles |
|---|---|
| electron | |
| neutron | |
| proton | |

[3]

**(b)** State why the formula for a magnesium ion is $Mg^{2+}$ rather than $Mg^{+}$ or $Mg^{3+}$.

.....

..... [1]

**(c)** The formula for a nitride ion is $N^{3-}$.

Deduce the formula for magnesium nitride.

..... [1]

**(d)** Magnesium nitrate, $Mg(NO_3)_2$, is an ionic compound.

Predict **two** physical properties of magnesium nitrate.

1 .....

2 .....
[2]

**(e)** Calculate the percentage by mass of nitrogen in magnesium nitrate.

Give your answer to **two** significant figures.

percentage by mass ..... [2]

[Total: 9]

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**5** Carbon reacts with steam to make carbon monoxide and hydrogen.

This reaction is reversible. The forward reaction absorbs thermal energy.

$$C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$$

**(a)** An equilibrium mixture is formed when the reversible reaction happens in a closed system.

**(i)** Explain why the reversible reaction must be in a closed system for an equilibrium mixture to be formed.

.....

..... [1]

**(ii)** Predict what happens to the **position of equilibrium** when the temperature is decreased and the pressure remains constant.

Explain your answer.

prediction .....

explanation .....

.....

..... [2]

**(iii)** Predict what happens to the **position of equilibrium** when the pressure is decreased and the temperature remains constant.

Explain your answer.

prediction .....

explanation .....

.....

..... [2]

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**(b)** Predict what happens to the **rate of the backward reaction** when the temperature is decreased and the pressure remains constant.

Explain your answer.

prediction .....

explanation .....

.....

.....

.....

[2]

**(c)** Predict what happens to the **rate of the backward reaction** when the pressure is increased and the temperature remains constant.

Explain your answer.

prediction .....

explanation .....

.....

.....

.....

[2]

[Total: 9]

**6** This question is about the energy changes that take place during chemical reactions.

**(a)** Nitrogen reacts with oxygen to make nitrogen monoxide.

The reaction is endothermic.

$$N_2(g) + O_2(g) \longrightarrow 2NO(g)$$

Draw, on the axes provided in Fig. 6.1, the reaction pathway diagram for this reaction.

Include labels for the:

- axes
- reactants
- product
- enthalpy change of reaction, $\Delta H$
- activation energy, $E_a$.

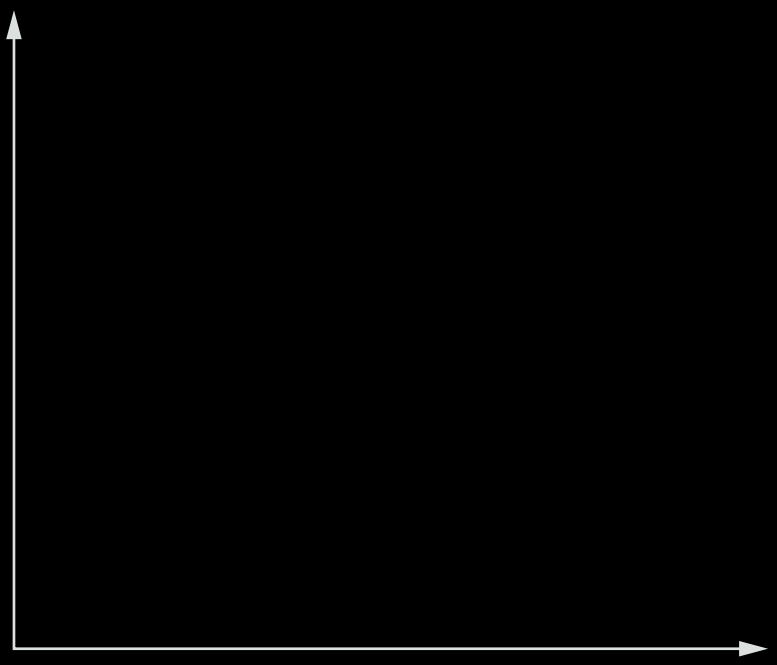


**Fig. 6.1**

[5]

**(b)** Hydrogen reacts with bromine to make hydrogen bromide.

$$H—H + Br—Br \rightarrow 2H—Br$$

Calculate the enthalpy change of this reaction.

Use the bond energies in Table 6.1.

**Table 6.1**

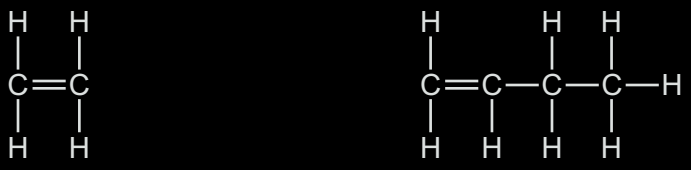
| bond | bond energy in kJ / mol |
|---|---|
| H—H | 436 |
| Br—Br | 193 |
| H—Br | 366 |

enthalpy change of reaction ..... kJ / mol [3]

[Total: 8]

**7** Ethene, but-1-ene and but-2-ene are alkenes.

The displayed formulae of ethene and but-1-ene are shown in Fig. 7.1.



ethene but-1-ene

**Fig. 7.1**

**(a)** State the general formula of the homologous series of alkenes.

..... [1]

**(b)** But-1-ene and but-2-ene are structural isomers.

**(i)** State the meaning of the term structural isomers.

.....

..... [1]

**(ii)** Draw the displayed formula for but-2-ene.

[1]

**(c)** But-1-ene is an unsaturated hydrocarbon.

**(i)** State why but-1-ene is an unsaturated compound.

.....

..... [1]

**(ii)** State why but-1-ene is a hydrocarbon.

.....

..... [1]

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**(d)** But-1-ene reacts with steam in the presence of a catalyst and reacts with bromine.

**(i)** Draw the structural formula of the product of the reaction with steam in the presence of a catalyst.

[1]

**(ii)** Draw the structural formula of the product of the reaction with bromine.

[1]

**(e)** Ethene is a covalent substance.

**(i)** Draw a dot-and-cross diagram to show the bonding in a molecule of ethene.

Include only the outer shell electrons of each atom.

[2]

**(ii)** Explain why ethene has a low melting point.

.....

..... [1]

[Total: 10]

**8** This question is about electrolysis.

**(a)** The table shows some information about the electrolysis of three different electrolytes using graphite electrodes.

Complete Table 8.1 with the names of the products at each electrode.

**Table 8.1**

| electrolyte | product at anode | product at cathode |
|---|---|---|
| concentrated aqueous potassium iodide | | |
| dilute sulfuric acid | | |
| molten lead(II) bromide | | |

[3]

**(b)** Give **two** reasons why graphite is a suitable material from which to make electrodes.

1 .....

2 .....

[2]

**(c)** Aluminium is manufactured by the electrolysis of aluminium oxide dissolved in molten cryolite.

At the anode, oxide ions react to make oxygen molecules.

Construct the ionic half-equation for this electrode reaction.

..... [1]

[Total: 6]

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**9** A sample of clean, dry air contains 0.0400% carbon dioxide by volume.

**(a)** Calculate the number of molecules of carbon dioxide in 480 $dm^3$ of clean, dry air at room temperature and pressure.

One mole of any gas contains $6.02 \times 10^{23}$ molecules.

number of molecules ..... [3]

**(b)** Complete combustion of fuels such as gasoline makes carbon dioxide.

One of the hydrocarbons in gasoline has the molecular formula $C_9H_{20}$.

Construct the equation for the complete combustion of $C_9H_{20}$.

..... [1]

**(c)** Higher levels of atmospheric carbon dioxide lead to increased global warming.

**(i)** Give **one** adverse effect of global warming.

..... [1]

**(ii)** Describe how the presence of gases such as carbon dioxide in the atmosphere causes global warming.

.....

.....

..... [2]

**(d)** Carbon dioxide is removed from the atmosphere by photosynthesis.

State the word equation for photosynthesis.

..... [1]

[Total: 8]

**10** Proteins and PET are polymers made by a reaction called condensation polymerisation.

**(a)** The diagram in Fig. 10.1 shows the structure of a section of a protein.

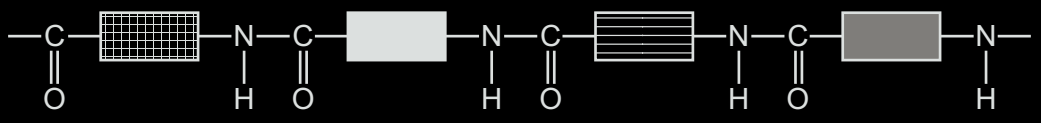


**Fig. 10.1**

**(i)** Draw the general structure of the amino acid monomers used to make proteins.

[1]

**(ii)** Proteins are polyamides.

Name one **other** polyamide.

..... [1]

**(b)** PET is a polymer used to make plastic bottles.

The diagram in Fig. 10.2 shows the structure of PET.

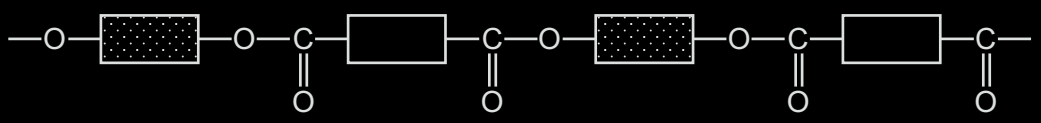


**Fig. 10.2**

**(i)** Name the linkage present in PET.

..... [1]

**(ii)** Proteins and PET are both made by condensation polymerisation.

Describe the differences between condensation polymerisation and addition polymerisation.

.....

.....

.....

..... [2]

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**(c)** Describe **two** environmental challenges caused by the disposal of plastics such as PET.

1 .....

.....

2 .....

.....

[2]

[Total: 7]

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## The Periodic Table of Elements

| Group | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| I | II | | | | | | | | | | | III | IV | V | VI | VII | VIII |
| | | | | | | | 1 H hydrogen 1 | | | | | | | | | | 2 He helium 4 |
| 3 Li lithium 7 | 4 Be beryllium 9 | | | | | | | | | | | 5 B boron 11 | 6 C carbon 12 | 7 N nitrogen 14 | 8 O oxygen 16 | 9 F fluorine 19 | 10 Ne neon 20 |
| 11 Na sodium 23 | 12 Mg magnesium 24 | | | | | | | | | | | 13 Al aluminium 27 | 14 Si silicon 28 | 15 P phosphorus 31 | 16 S sulfur 32 | 17 Cl chlorine 35.5 | 18 Ar argon 40 |
| 19 K potassium 39 | 20 Ca calcium 40 | 21 Sc scandium 45 | 22 Ti titanium 48 | 23 V vanadium 51 | 24 Cr chromium 52 | 25 Mn manganese 55 | 26 Fe iron 56 | 27 Co cobalt 59 | 28 Ni nickel 59 | 29 Cu copper 64 | 30 Zn zinc 65 | 31 Ga gallium 70 | 32 Ge germanium 73 | 33 As arsenic 75 | 34 Se selenium 79 | 35 Br bromine 80 | 36 Kr krypton 84 |
| 37 Rb rubidium 85 | 38 Sr strontium 88 | 39 Y yttrium 89 | 40 Zr zirconium 91 | 41 Nb niobium 93 | 42 Mo molybdenum 96 | 43 Tc technetium – | 44 Ru ruthenium 101 | 45 Rh rhodium 103 | 46 Pd palladium 106 | 47 Ag silver 108 | 48 Cd cadmium 112 | 49 In indium 115 | 50 Sn tin 119 | 51 Sb antimony 122 | 52 Te tellurium 128 | 53 I iodine 127 | 54 Xe xenon 131 |
| 55 Cs caesium 133 | 56 Ba barium 137 | 57–71 lanthanoids | 72 Hf hafnium 178 | 73 Ta tantalum 181 | 74 W tungsten 184 | 75 Re rhenium 186 | 76 Os osmium 190 | 77 Ir iridium 192 | 78 Pt platinum 195 | 79 Au gold 197 | 80 Hg mercury 201 | 81 Tl thallium 204 | 82 Pb lead 207 | 83 Bi bismuth 209 | 84 Po polonium – | 85 At astatine – | 86 Rn radon – |
| 87 Fr francium – | 88 Ra radium – | 89–103 actinoids | 104 Rf rutherfordium – | 105 Db dubnium – | 106 Sg seaborgium – | 107 Bh bohrium – | 108 Hs hassium – | 109 Mt meitnerium – | 110 Ds darmstadtium – | 111 Rg roentgenium – | 112 Cn copernicium – | 113 Nh nihonium – | 114 Fl flerovium – | 115 Mc moscovium – | 116 Lv livermorium – | 117 Ts tennessine – | 118 Og oganesson – |

**Key**

atomic number
atomic symbol
name
relative atomic mass

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| lanthanoids | 57 La lanthanum 139 | 58 Ce cerium 140 | 59 Pr praseodymium 141 | 60 Nd neodymium 144 | 61 Pm promethium – | 62 Sm samarium 150 | 63 Eu europium 152 | 64 Gd gadolinium 157 | 65 Tb terbium 159 | 66 Dy dysprosium 163 | 67 Ho holmium 165 | 68 Er erbium 167 | 69 Tm thulium 169 | 70 Yb ytterbium 173 | 71 Lu lutetium 175 |
| actinoids | 89 Ac actinium – | 90 Th thorium 232 | 91 Pa protactinium 231 | 92 U uranium 238 | 93 Np neptunium – | 94 Pu plutonium – | 95 Am americium – | 96 Cm curium – | 97 Bk berkelium – | 98 Cf californium – | 99 Es einsteinium – | 100 Fm fermium – | 101 Md mendelevium – | 102 No nobelium – | 103 Lr lawrencium – |

The volume of one mole of any gas is 24 $dm^3$ at room temperature and pressure (r.t.p.).